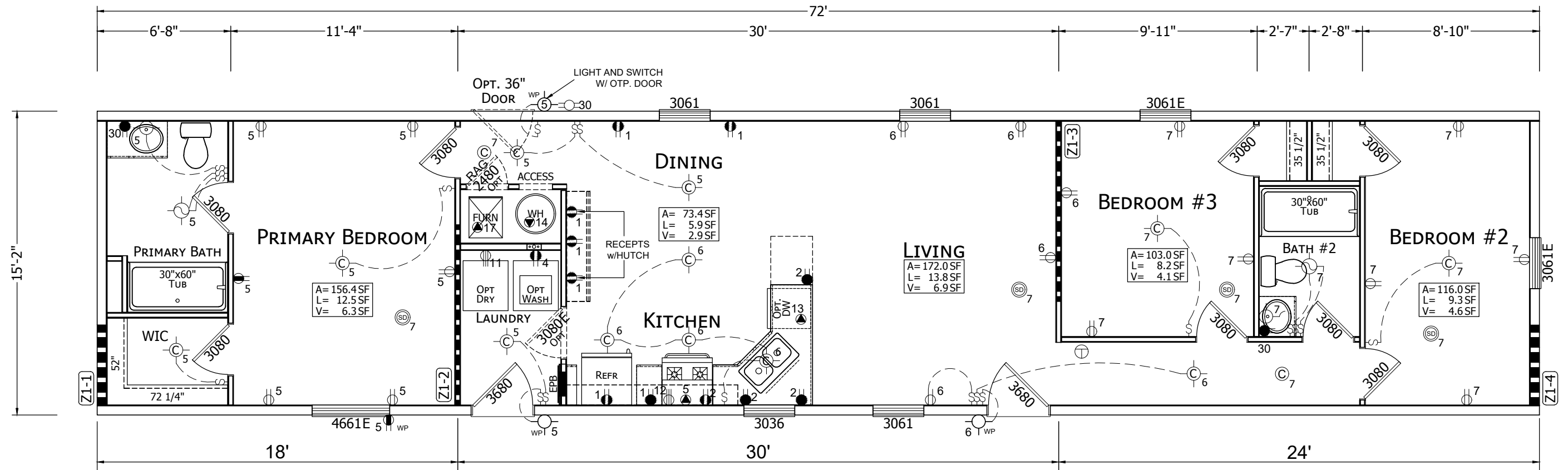




OPT. REVERSE AISLE

DESIGN CRITERIA

ZONES: WIND 1, ROOF MIDDLE, TEMP III

TOTAL ROUGH WINDOW OPENINGS: 92.6 SF

CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

1 2025 CODE UPDATE

2/14/2025 JR

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: 3/16" = 1'-0"

TITLE:
FLOOR PLAN
ELECTRICAL PLAN

FILENAME: 112AP-1672H32087

SHEET:
AE-103

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Overall Check = PASS

CHB Shear Wall - Layout / Design

Model#: 112AP-1672H32087opt2bed

Wind Zone = I

Wind Load = 15 psf

Diaphragm Construction:

Diaphragm Type = Gypsum Ceiling

Foam Brand = F-2100

Bottom Plate Thickness = 2x

Top Plate Thickness = 2x

Wall Construction:

Ground Tie Chart

Module Width = 182 in

Number of Boxes = 1

Wall Height = 96 in

Roof Pitch = 4.36 :12

Angle = 20.0 °

Roof Overhang = 4 in

Floor Joist Spacing = 16 in o.c.

Rim Joist Plies = 1

Frame Spacing = 99.5 in

Beam Depth = 12 in

Open Porch = 0.00 ft.

Home Length = 72.00 ft.

Minimum Width = 182 in

Max plf = 228 plf

Max. End Wall Reaction = 3,458 lbs

Max. Interior Reaction = 6,916 lbs

Max. Span = 48.00 ft.

Floor Construction:

Floor Joist Size = 2x8

Floor Joist Species = SPF

Floor Joist Grade = #3

Decking Glued to Joists = Yes

Peir Height (in.)

Near Beam Angle (Deg.)

Far Beam Angle (Deg.)

Ground Tie Spacing (ft)

Max. Spacing (ft.)

12

20

28

36

44

52

60

68

32.79

40.66

47.04

52.19

56.37

59.80

62.64

65.03

13.08

11.80

10.60

9.54

8.62

7.82

12.60

12.68

13'-0"

11'-9"

10'-7"

9'-6"

8'-7"

7'-9"

12'-7"

12'-8"

* For monoslope homes only

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Vertical Trib = 4.00 ft

King Post Height *= in.

Porch End Wall = --

Left End Wall Shear Wall

Dist. To Next SW = 18.00 ft.

Manual Input = --

Load to Shearwall = 540.0 lbs

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 48.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 135.0 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

*Construct per Note 6 on SW01.35.XX

Zone - ID

Z1-1

Joist

2

Wall Conn.

8

SC O.C.

6 @ 8"

SF O.C.

10 @ 5"

Beam Lag

1

*May be 1 joist per Note 9.C. on SW01.00.04

Result = PASS

Shear Wall 2

Dist. To Next SW = 32.83 ft.

Manual Input = --

Load to Shearwall = 1524.9 lbs

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 108.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 169.4 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

*Construct per Note 6 on SW01.35.XX

Zone - ID

Z1-2

Joist

2

Wall Conn.

8

SC O.C.

16 @ 7"

SF O.C.

18 @ 6"

Beam Lag

2

Result = PASS

Shear Wall 3

Dist. To Next SW = 21.16 ft.

Manual Input = --

Load to Shearwall = 1619.7 lbs

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 108.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 180.0 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

*Construct per Note 6 on SW01.35.XX

Zone - ID

Z1-3

Joist

2

Wall Conn.

8

SC O.C.

18 @ 6"

SF O.C.

20 @ 5.5"

Beam Lag

2

Result = PASS

Shear Wall 4

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 634.8 lbs

Panel Layout

From Module Edge = 6.00 in.

Panel Length = 48.00 in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = 158.7 plf

Shearwall Panel = Gypsum

Gyp. Brand = U.S.G.

Fasteners = Yes

Adhesive = PVA

One Side or Two = 1

Max. Shearwall plf = 247 plf

*Construct per Note 6 on SW01.35.XX

Zone - ID

Z1-4

Joist

2

Wall Conn.

8

SC O.C.

8 @ 6"

SF O.C.

12 @ 4"

Beam Lag

1

*May be 1 joist per Note 9.C on SW01.00.04

Result = PASS

Shear Wall 5

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =

Shear Wall 6

Dist. To Next SW = ft.

Manual Input = --

Load to Shearwall = 0.0 lbs

Panel Layout

From Module Edge = in.

Panel Length = in.

From Module Edge = in.

Panel Length = in.

Shearwall plf = N/A plf

Shearwall Panel =

Gyp. Brand =

Fasteners =

Adhesive =

One Side or Two =

Max. Shearwall plf = plf

*Construct per SW01.25.01

Zone - ID

Joist

Wall Conn.

SC O.C.

SF O.C.

Beam Lag

Result =



MODIFICATIONS

PROJECT: 112AP-1672H32087
3 BD 2 BTH

DRAWN BY: J. RILEY

DATE: 09/18/23

SCALE: NTS

TITLE: SHEAR WALLS
OPT 2 BEDROOM

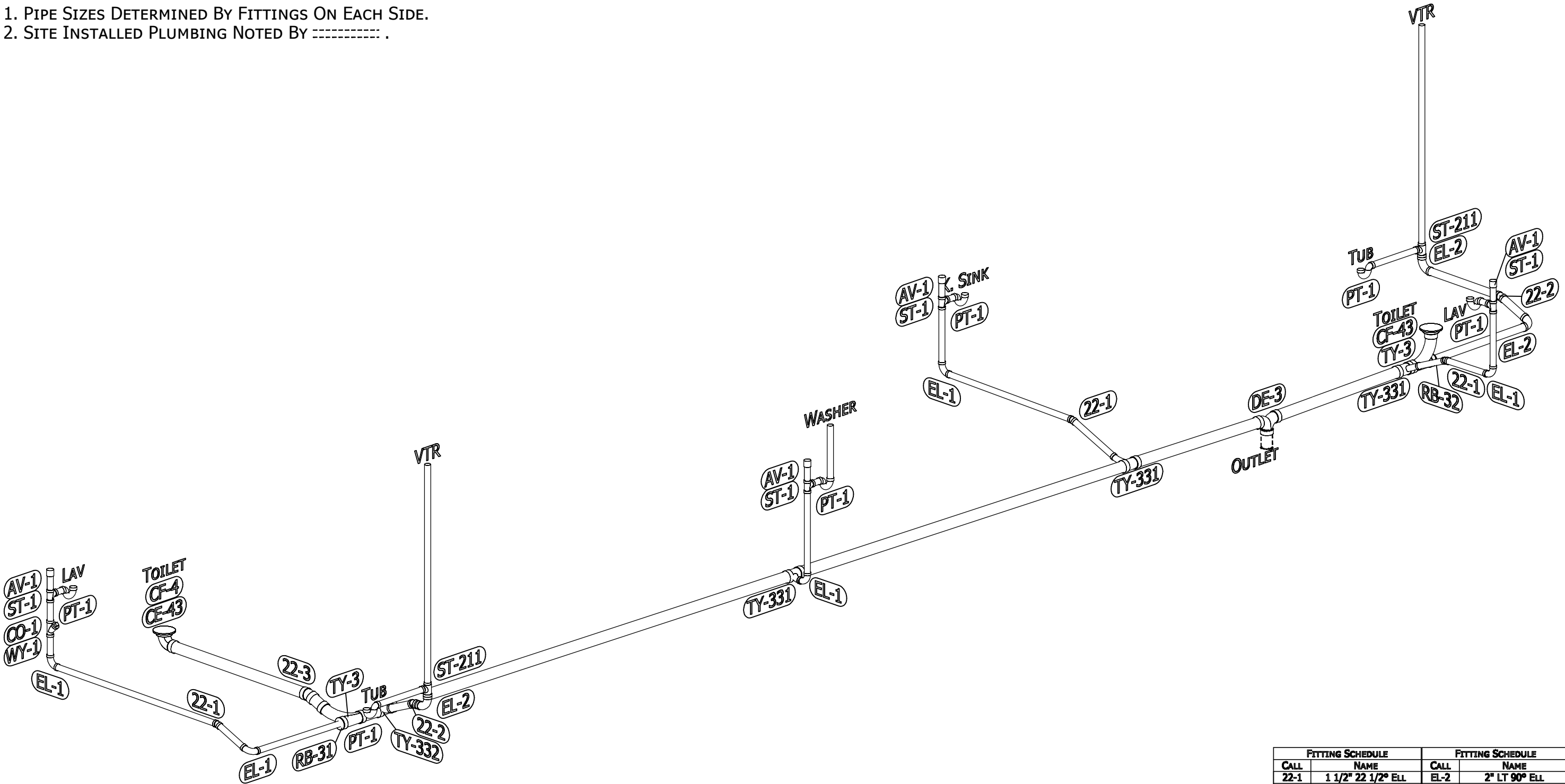
FILENAME: 112AP-1672H32087

SHEET: SW-102

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NOTES:

- 1. PIPE SIZES DETERMINED BY FITTINGS ON EACH SIDE.
- 2. SITE INSTALLED PLUMBING NOTED BY ----- .



FITTING SCHEDULE		FITTING SCHEDULE	
CALL	NAME	CALL	NAME
22-1	1 1/2" 22 1/2° ELL	EL-2	2" LT 90° ELL
22-2	2" 22 1/2° ELL	PT-1	1 1/2" P-TRAP
22-3	3" 22 1/2° ELL	RB-31	3" x 1 1/2" BUSHING
AV-1	1 1/2" AUTO VENT	RB-32	3" x 2" BUSHING
CE-43	4" x 3" CLOSET ELL	ST-1	1 1/2" TEE
CF-4	4" CLOSET FLANGE	ST-211	2" x 1 1/2" x 1 1/2" TEE
CF-43	4" x 3" CLOSET FLANGE	TY-3	3" LTTY
CO-1	1 1/2" CLEANOUT	TY-331	3" x 3" x 1 1/2" LTTY
DE-3	3" DOUBLE ELL	TY-332	3" x 3" x 2" LTTY
EL-1	1 1/2" LT 90° ELL	WY-1	1 1/2" WYE

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8900

PFS^{#23}

Conforms to
HUD MHCSS
9/25/23

MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
DRAIN LINE
PLAN

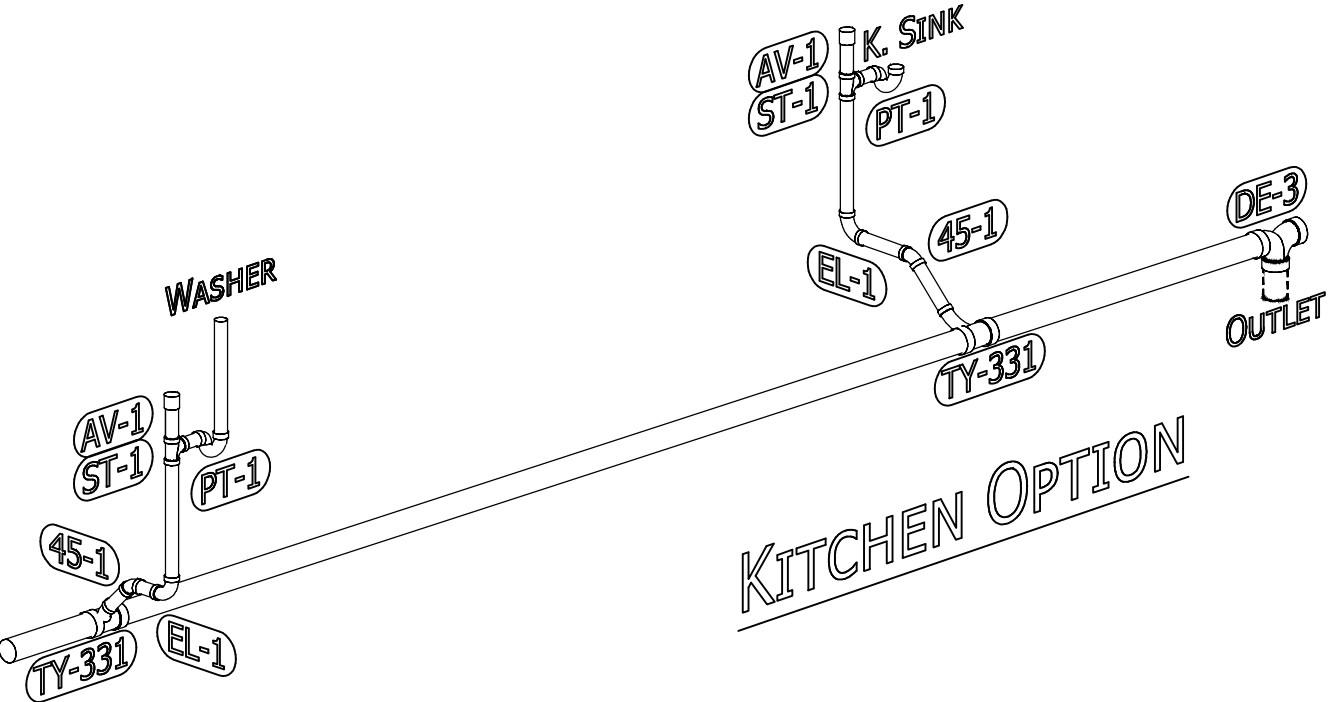
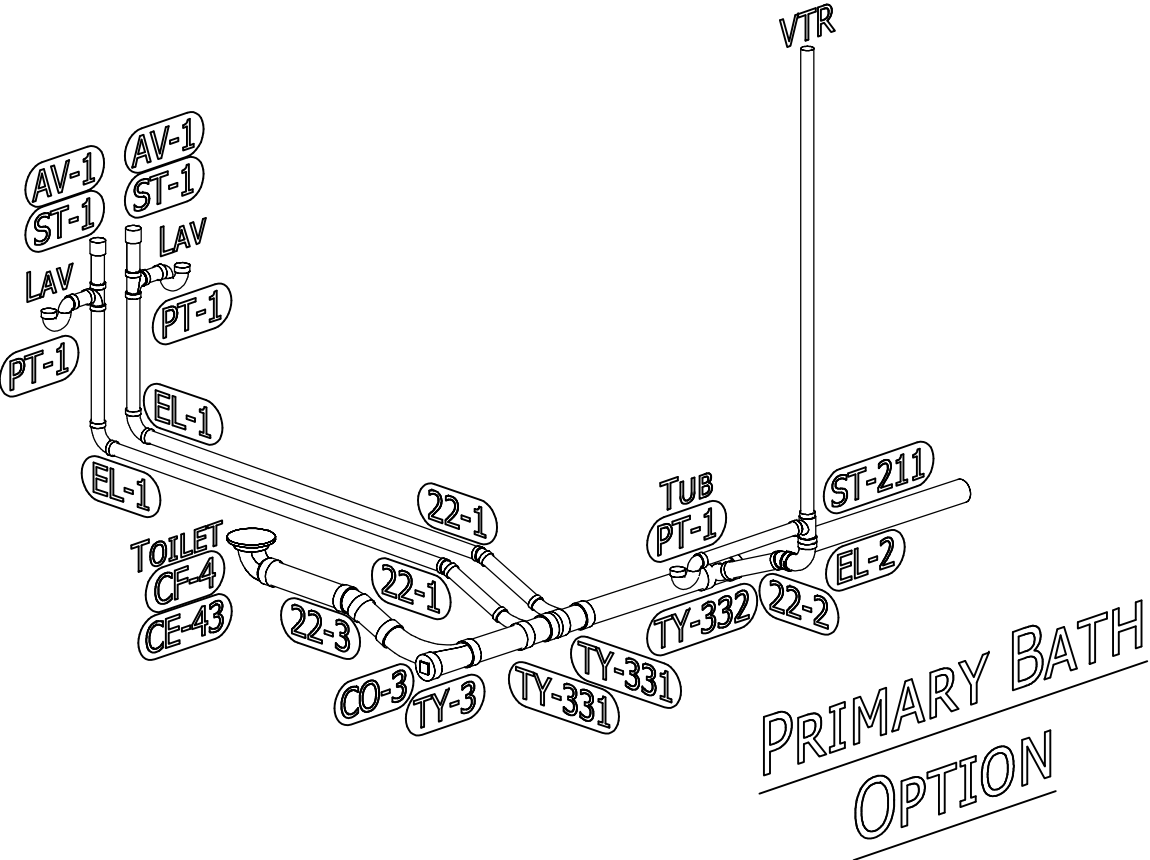
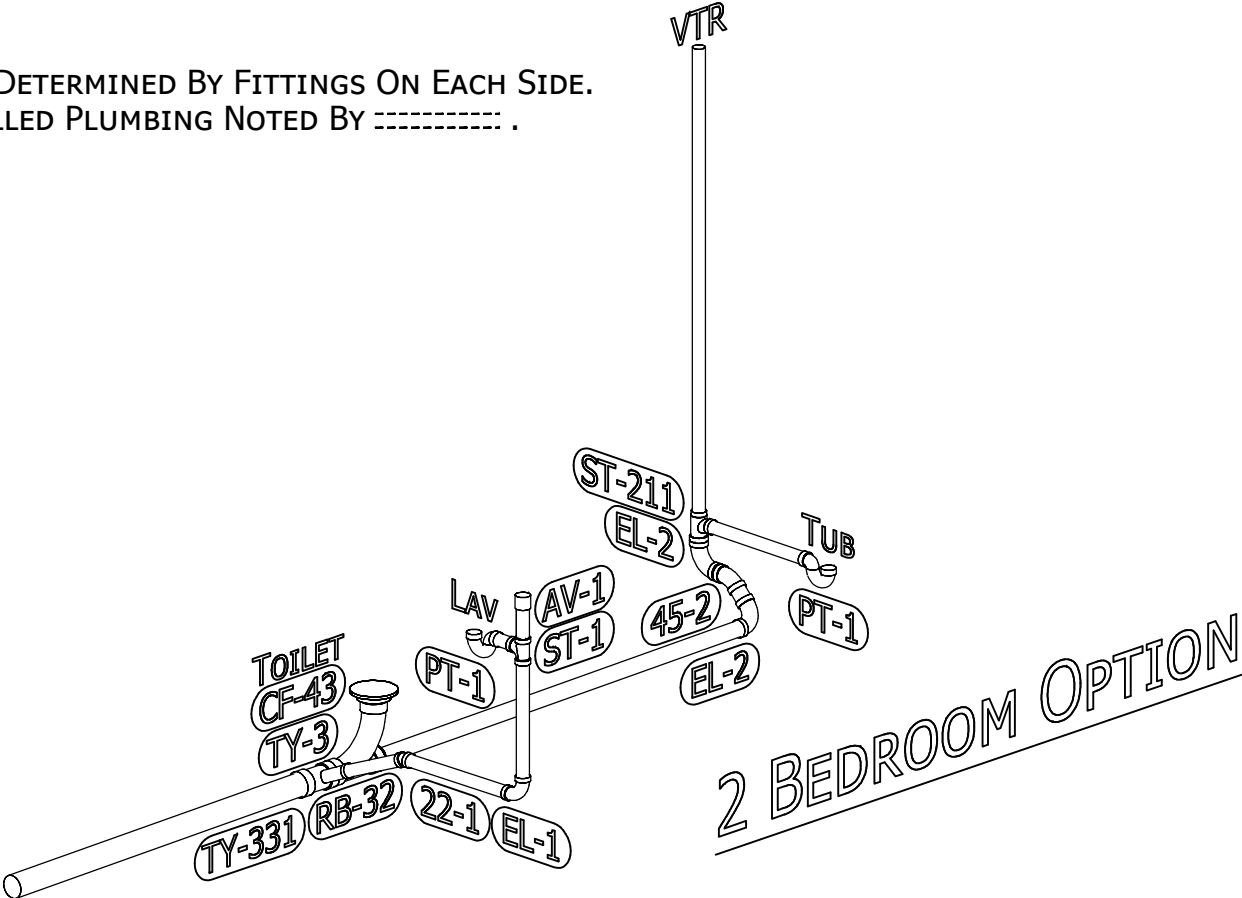
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SHEET:
D-101

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FITTING SCHEDULE		FITTING SCHEDULE	
CALL	NAME	CALL	NAME
22-1	1 1/2" 22 1/2° ELL	DE-3	3" DOUBLE ELL
22-2	2" 22 1/2° ELL	EL-1	1 1/2" LT 90° ELL
22-3	3" 22 1/2° ELL	EL-2	2" LT 90° ELL
45-1	1 1/2" 45° ELL	PT-1	1 1/2" P-TRAP
45-2	2" 45° ELL	RB-32	3" x 2" BUSHING
AV-1	1 1/2" AUTO VENT	ST-1	1 1/2" TEE
CE-43	4" x 3" CLOSET ELL	ST-211	2" x 1 1/2" x 1 1/2" TEE
CF-4	4" CLOSET FLANGE	TY-3	3" LTTY
CF-43	4" x 3" CLOSET FLANGE	TY-331	3" x 3" x 1 1/2" LTTY
CO-3	3" CLEANOUT	TY-332	3" x 3" x 2" LTTY

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8900

PFS#23

Conforms to
HUD MHCSS
9/25/23

MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
DRAIN LINE
PLAN

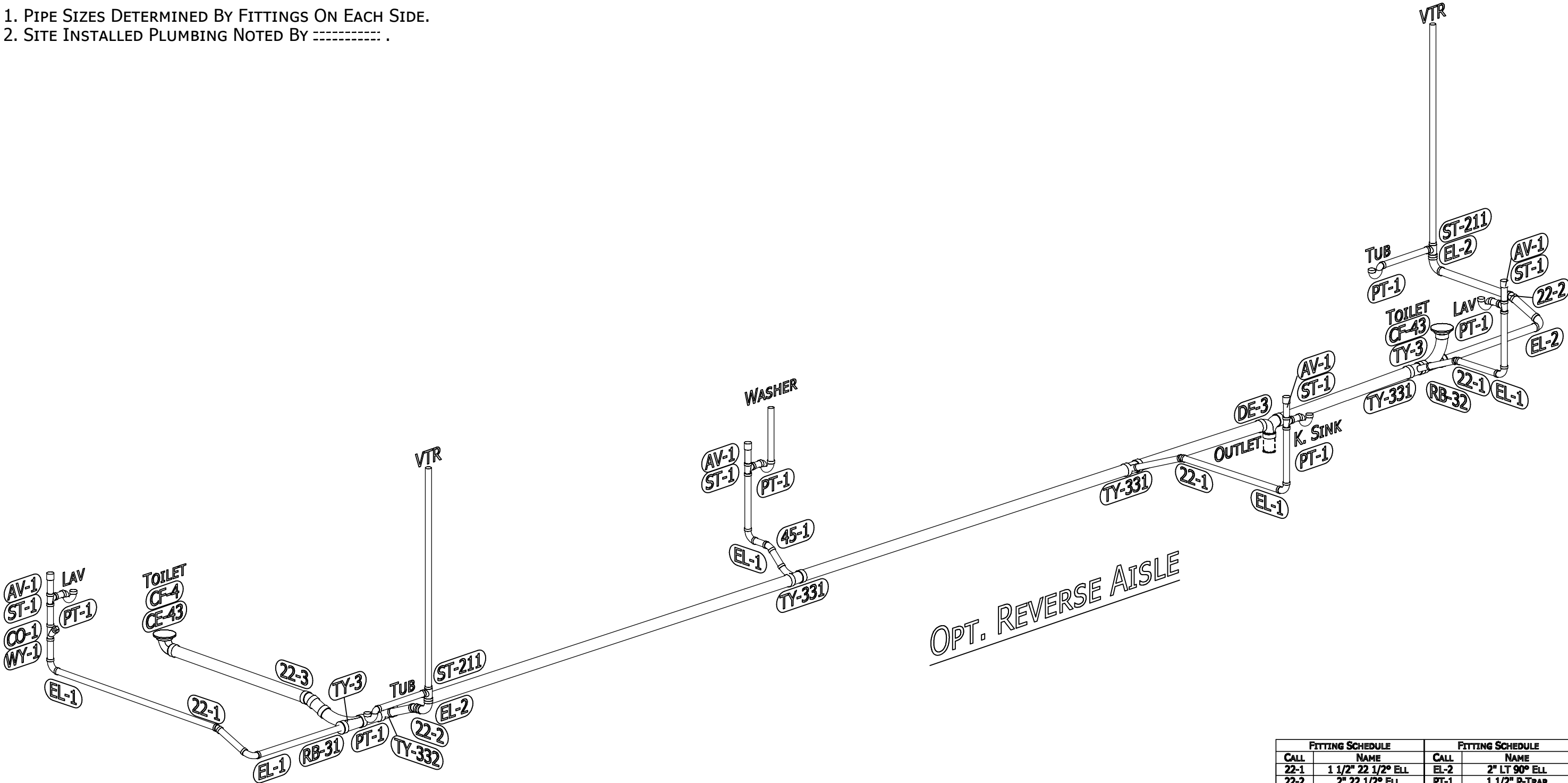
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SHEET:
D-102

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NOTES:

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- 2. SITE INSTALLED PLUMBING NOTED BY ----- .



FITTING SCHEDULE		FITTING SCHEDULE	
CALL	NAME	CALL	NAME
22-1	1 1/2" 22 1/2° ELL	EL-2	2" LT 90° ELL
22-2	2" 22 1/2° ELL	PT-1	1 1/2" P-TRAP
22-3	3" 22 1/2° ELL	RB-31	3" x 1 1/2" BUSHING
45-1	1 1/2" 45° ELL	RB-32	3" x 2" BUSHING
AV-1	1 1/2" AUTO VENT	ST-1	1 1/2" TEE
CE-43	4" x 3" CLOSET ELL	ST-211	2" x 1 1/2" x 1 1/2" TEE
CF-4	4" CLOSET FLANGE	TY-3	3" LT TY
CF-43	4" x 3" CLOSET FLANGE	TY-331	3" x 3" x 1 1/2" LT TY
CO-1	1 1/2" CLEANOUT	TY-332	3" x 3" x 2" LT TY
DE-3	3" DOUBLE ELL	WY-1	1 1/2" WYE
EL-1	1 1/2" LT 90° ELL		

CHAMPION

HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8900

PFS^{#23}

Conforms to
HUD MHCSS
9/25/23

MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
DRAIN LINE
PLAN

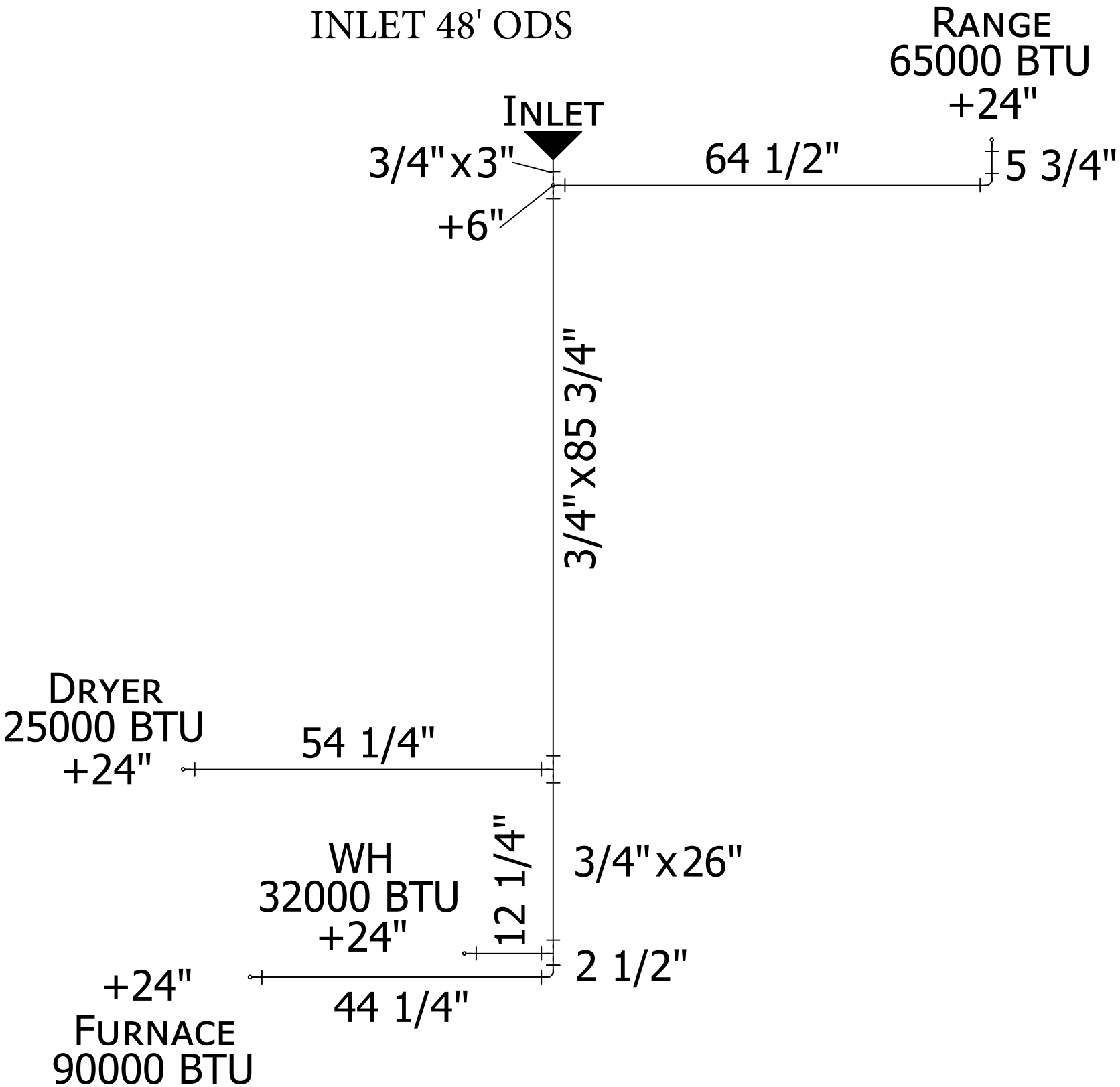
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SHEET:
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NOTES:

- 1. ALL PIPE SIZES 1/2" UNLESS OTHERWISE SPECIFIED
- 2. TOTAL BTU = 212,000
- 3. COLUMN LENGTH = 20'-0"



CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
GAS LINE
PLAN

FILENAME: 112AP-1672H32087

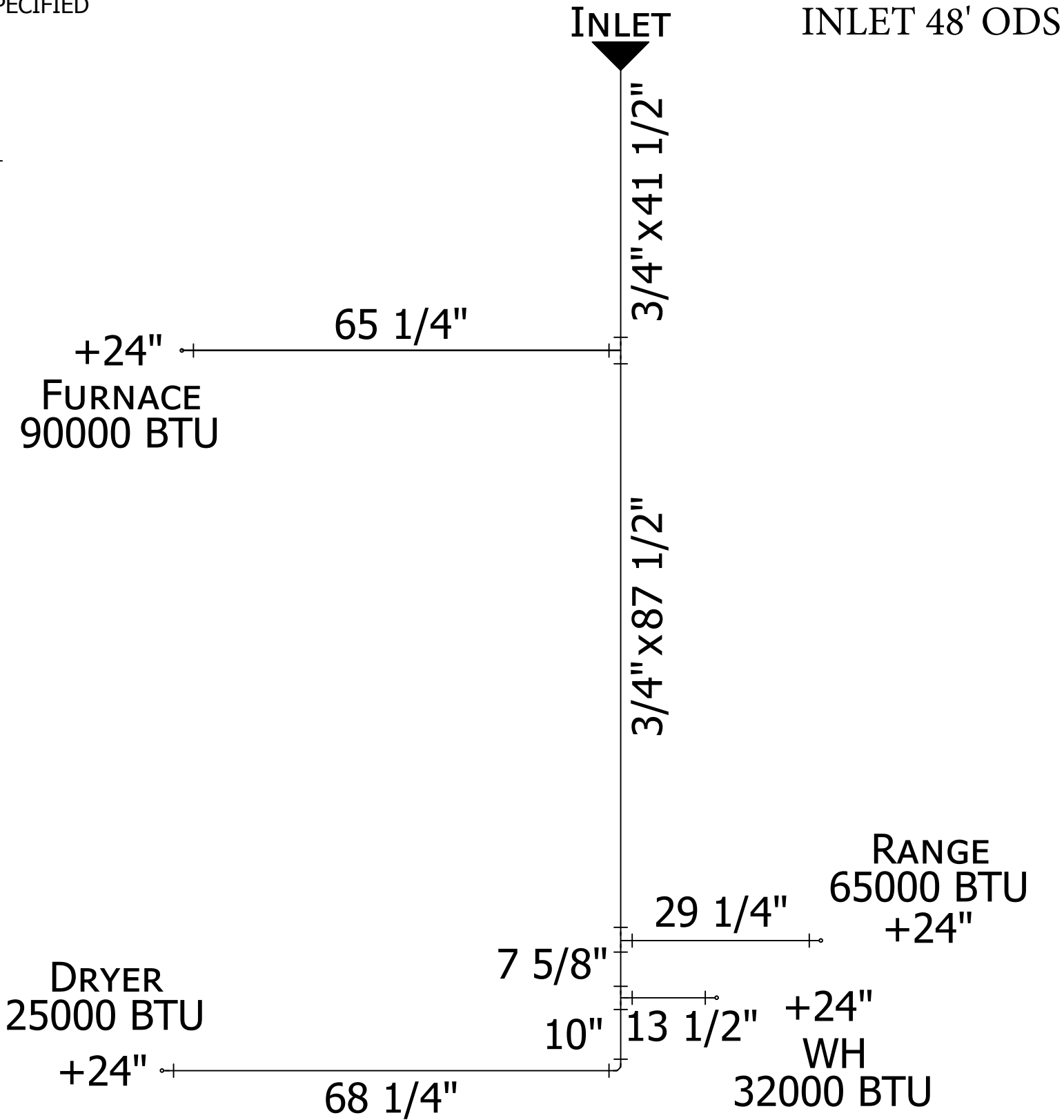
SHEET:
G-101

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NOTES:

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- 2. TOTAL BTU = 212,000
- 3. COLUMN LENGTH = 20'-0"

KITCHEN OPTION



CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8200



MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
GAS LINE
PLAN

FILENAME: 112AP-1672H32087

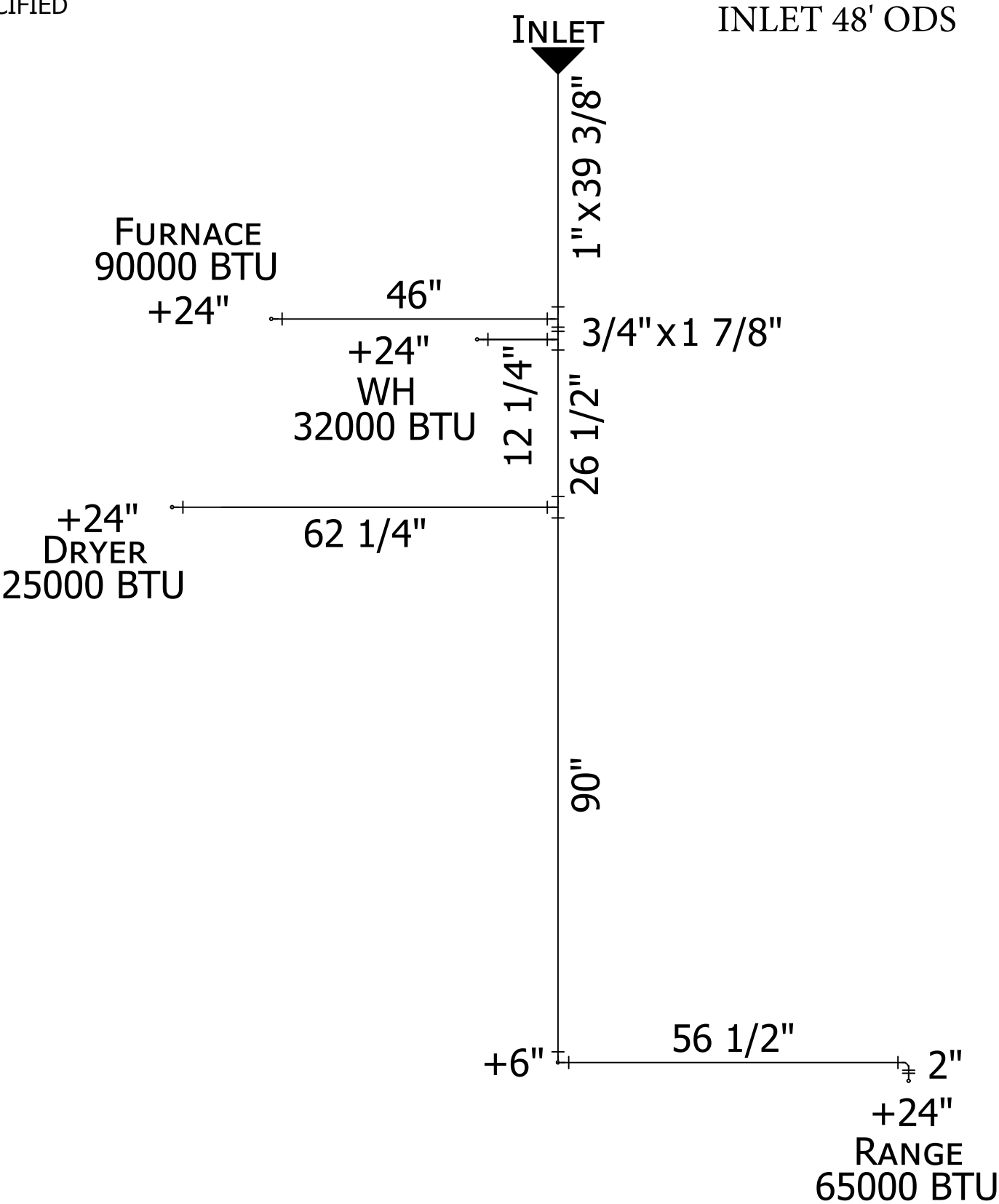
SHEET:
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NOTES:

- 1. ALL PIPE SIZES 1/2" UNLESS OTHERWISE SPECIFIED
- 2. TOTAL BTU = 212,000
- 3. COLUMN LENGTH = 30'-0"

OPT. REVERSE AISLE



CHAMPION
HOME BUILDERS

755 W. BIG BEAVER ROAD, SUITE 1000 TROY, MI 48064
PHONE: 248-614-8900



MODIFICATIONS

PROJECT:
112AP-1672H32087
3 BD 2 BTH

DRAWN BY: A.GOULD
DATE: 08-21-23
SCALE: NTS

TITLE:
GAS LINE
PLAN

FILENAME: 112AP-1672H32087

SHEET:
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